

Solution Code - Module 3-Core Java - TryCatchExample

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q20_TryCatchExample.java

```
1 | import java.util.Scanner;
2 |
3 | public class Q20_TryCatchExample {
4 |     public static void main(String[] args) {
5 |         Scanner scanner = new Scanner(System.in);
6 |         System.out.println("=== Division Handler ===");
7 |
8 |         try {
9 |             System.out.print("Enter the first integer (numerator): ");
10 |             int numerator = scanner.nextInt();
11 |
12 |             System.out.print("Enter the second integer (denominator): ");
13 |             int denominator = scanner.nextInt();
14 |
15 |             // This operation can throw ArithmeticException
16 |             int result = numerator / denominator;
17 |             System.out.println("Result of division: " + result);
18 |
19 |         } catch (ArithmeticException e) {
20 |             System.out.println("Error caught: ArithmeticException - Division by zero is not allowed.");
21 |         } catch (Exception e) {
22 |             System.out.println("Error caught: " + e.getMessage() + ". Please enter valid integers.");
23 |         } finally {
24 |             scanner.close();
25 |         }
26 |     }
27 | }
```